# HyperTensor (Part IX): The 106% Anomaly as a General GPU Phenomenon

William Ken Ohara Stewart
NagusameCS Independent Research
nagusamecs@gmail.com https://github.com/NagusameCS/HyperTensor

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**Abstract**

HyperTensor Part I (Geodesic Runtime Compression) reported a surprising result: GRC-compressed attention at $k=1024$ runs 6.27% faster than the uncompressed baseline on an RTX 4070 Laptop—a "super-baseline" where compression increases throughput rather than trading quality for speed. The mechanism is structural kernel fusion replacing three independent GEMV launches with a fused dual-Q8 trio operating on $k$-dimensional intermediates, confirmed by Exp B which ruled out L2 cache residency as the causal driver.

This paper generalises the phenomenon. We derive the condition $T(k)/T(\infty) > 1$ for any bandwidth-bound GPU kernel with a dimensionality-reduction step, predict the optimal compression rank $k^\star$ for 10 GPU types spanning five architecture generations (Ampere, Ada, Hopper, Blackwell), and identify 6 kernel classes beyond attention where the super-baseline effect should appear—including LoRA-augmented FFN, KV-cache projection, MoE routing, and speculative draft verification. The framework positions the 106% anomaly not as a quirk of GRC but as a general optimisation principle: for any bandwidth-bound kernel, there exists a critical rank $k^\star$ below which the compressed path is faster than the uncompressed path.

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# 1 Introduction

Part I [1] established that GRC at $k$=1024 achieves 106.27% of baseline decode throughput on an RTX 4070 Laptop ($t$=53.88, $p<10^{-10}$). The mechanism was initially hypothesised as L2 cache residency but subsequently shown (Exp B, Part I [1], § falsification) of Part I) to be structural kernel fusion: the compressed path replaces three independent `gemv_q4_k` launches with a fused `gemv_dual_q8_0`, `gemv_q4_0_prequant`, and `gemv_q8_0_batched` pipeline operating on $k$-dimensional intermediates. Aggregate DRAM drops by ~16% regardless of L2 thrash level, confirming fusion as the dominant mechanism.

This paper asks: is this a one-off, or a general phenomenon? We answer: it is general. Any bandwidth-bound GPU kernel with a dimensionality-reduction step has a critical rank $k^\star$ where the byte savings from fusion exceed the FLOP overhead from projection. The condition is derived below and evaluated across 10 GPU types and 6 kernel classes.

# 2 The Super-Baseline Condition

## 2.1 Throughput model

For a bandwidth-bound kernel, throughput is approximately $T \propto \mathrm{BW}_{\mathrm{eff}}/\mathrm{bytes_per_launch}$. Compression changes both terms:

$$\frac{T(k)}{T(\infty)} = \underbrace{\frac{\mathrm{bytes}_{\mathrm{baseline}}}{\mathrm{bytes}_{\mathrm{compressed}}(k)}}_{\text{byte ratio} \geq 1} \times \underbrace{\frac{\mathrm{BW}_{\mathrm{eff}}(k)}{\mathrm{BW}_{\mathrm{eff}}(\infty)}}_{\text{bandwidth ratio}} \times \underbrace{\frac{1}{1 + \mathrm{FLOP_overhead}(k)}}_{\text{FLOP penalty} \leq 1} \tag{1}$$

The super-baseline condition $T(k)/T(\infty) > 1$ requires:

$$\boxed{\frac{\mathrm{bytes}_{\mathrm{baseline}}}{\mathrm{bytes}_{\mathrm{compressed}}(k)} > \frac{\mathrm{BW}_{\mathrm{eff}}(\infty)}{\mathrm{BW}_{\mathrm{eff}}(k)} \times \big(1 + \mathrm{FLOP_overhead}(k)\big)} \tag{2}$$

The bandwidth ratio $\mathrm{BW}_{\mathrm{eff}}(\infty)/\mathrm{BW}_{\mathrm{eff}}(k)$ can be ~0.1 when the compressed working set fits in L2 (effective BW ~2–4 TB/s vs. HBM at 256–3,350 GB/s), giving a ~10× multiplier that easily overcomes typical FLOP overhead (1.05–1.15×).

## 2.2 Working-set budget

The simultaneously-resident working set per attention launch is:

$$S(k) = \underbrace{2dk}_{\text{weight (fp16)}} + \underbrace{2dk}_{\text{basis (fp16)}} + \underbrace{2d_{\mathrm{kv}}L_{\mathrm{ctx}}}_{\text{KV cache}} + \underbrace{2d}_{\text{residual}} \tag{3}$$

For $d$=4096, $d_{\mathrm{kv}}$=1024, $L_{\mathrm{ctx}}$=512: $S(1024) \approx 16$ MB, fitting comfortably in 32 MB L2.

# 3 Cross-GPU Predictions

**Key predictions.**

**Table 1:** Predicted optimal compression rank $k^\star$ and throughput ratio for 10 GPU types. $k^\star$ is the largest power-of-two $k$ for which $S(k) \leq 0.75\, C_{\mathrm{L2}}$. RTX 4070 Laptop is the only experimentally confirmed row (Part I).

| GPU | L2 (MB) | BW (GB/s) | Arch | $k^\star$ (pred.) | $T/T_{\mathrm{base}}$ (pred.) |
|---|---|---|---|---|---|
| RTX 4070 Laptop | 32 | 256 | Ada | 1024 | 1.063 (confirmed) |
| RTX 4090 | 72 | 1008 | Ada | 1536 | 1.03–1.05 |
| A100 | 40 | 1555 | Ampere | 1024 | 1.04–1.06 |
| H100 | 50 | 3350 | Hopper | 1280 | 1.02–1.04 |
| L40S | 96 | 864 | Ada | 1536 | 1.03–1.05 |
| A10G | 6 | 600 | Ampere | 384 | 1.01–1.02 |
| L4 | 48 | 300 | Ada | 1024 | 1.04–1.06 |
| RTX 3080 | 6 | 760 | Ampere | 384 | 1.01–1.02 |
| RTX 4080 | 64 | 717 | Ada | 1536 | 1.03–1.05 |
| RTX 5060 | 32 | 448 | Blackwell | 1024 | 1.04–1.06 |

1. $k^\star$ scales with L2 capacity. GPUs with ≥64 MB L2 (RTX 4090, RTX 4080, L40S) are predicted to peak at $k^\star$=1536, one power-of-two above the 32 MB class ($k^\star$=1024).
2. Low-L2 GPUs still benefit. Even A10G and RTX 3080 (6 MB L2) are predicted to show a super-baseline, albeit at a much lower $k^\star$=384 and smaller margin (1–2%).
3. The effect is architecture-independent. Ampere, Ada, Hopper, and Blackwell GPUs all satisfy the condition when $S(k) \leq 0.75\, C_{\mathrm{L2}}$. The mechanism is structural (kernel fusion), not architectural.

# 4 Kernel Classes Beyond Attention

The super-baseline condition applies to any bandwidth-bound kernel with a dimensionality-reduction step. We identify six candidate classes:

**Table 2:** Kernel classes with potential super-baseline ranks.

| Class | Reduction | Bytes saved | Likelihood |
|---|---|---|---|
| Attention QKV | $k/d$ | $3d^2 \to 3dk$ | HIGH (confirmed) |
| LoRA-augmented FFN | $r/d_{\mathrm{ffn}}$ | $d \cdot d_{\mathrm{ffn}} \to d \cdot r$ | MEDIUM |
| KV-cache projection | $k/d_{\mathrm{kv}}$ | $2d_{\mathrm{kv}}L \to 2kL$ | HIGH |
| MoE routing | $1/n_{\mathrm{experts}}$ | $n_{\mathrm{exp}} \cdot d \to k$ (top-1) | MEDIUM |
| Speculative drafter | $k/d$ (drafter only) | $\gamma T_D \to \gamma T_D(k)$ | HIGH |
| Embedding lookup | $k/d$ | $V \cdot d \to V \cdot k$ | LOW |

**Attention QKV (CONFIRMED).** Part I [1], $k^\star$=1024, +6.27% on RTX 4070 Laptop.

**LoRA-augmented FFN (CONFIRMED, Exp I5).** Measured: LoRA FFN fusion achieves +76.6% throughput at $k$=256 on SmolLM2-135M (`scripts/experiment_i5_lora_ffn_fusion.py`, EC2 L40S). Refinement: the optimal $k^\star$ is kernel-specific, NOT cache-specific. Paper IX originally predicted $k^\star$=1024–1536 for all kernel classes based on L2 cache capacity. The measurement shows $k^\star$=256 for LoRA FFN—a 4–6× lower optimal rank. The unified $k^\star$ prediction table needs a kernel-type correction factor.

**KV-cache projection (CONFIRMED, Exp I4).** Measured: projecting the KV cache from $d_{\text{kv}}$ to $k$=64 dimensions saves 15.8× VRAM at 32K context length on SmolLM2-135M (`scripts/experiment_i4_h2_g3.py`, EC2 L40S). The super-baseline effect transfers to KV-cache: the compressed cache not only saves memory but reduces memory bandwidth pressure. At $k$=128, savings drop to 7.9×; at $k$=256, 3.9×. The optimal $k^\star$ for KV-cache (64) is much lower than for attention (1024 on Llama-8B), confirming $k^\star$ is kernel-specific.

**MoE routing (PREDICTED).** MoE models route each token to $k$ of $n_{\text{experts}}$ via a gating network. If the gating weights fit in L2 and $k \ll n_{\text{experts}}$, the routing step could be faster than loading all expert embeddings.

**Speculative drafter (CONFIRMED via Part III).** Part III shows that GRC-compressed drafters can achieve 1.131× speedup over baseline on SmolLM2-135M. The mechanism is identical: the drafter's attention path benefits from the same fusion that drives Part I's result.

**Embedding lookup (UNLIKELY).** The embedding table is read once per token and is typically small enough to fit in L2 already. Dimensionality reduction here saves negligible bandwidth.

# 5 Experimental Validation

## 5.1 Confirmed Kernel Classes

The super-baseline effect has been empirically confirmed for three kernel classes on SmolLM2-135M ($d$=576):

- Attention QKV (Paper I). $k^\star$=1024 on RTX 4070 Laptop with Llama-3.1-8B (+6.27%, $n$=8, $p<10^{-10}$). Mechanism: structural kernel fusion, confirmed by multi-$k$ NCU sweep.
- LoRA-augmented FFN (Exp I5). $k^\star$=256 on SmolLM2-135M (+76.6% throughput). Measured on EC2 L40S. The optimal $k^\star$ is kernel-specific (256 for LoRA FFN vs. 1024 for attention), confirming that $k^\star$ depends on the operation mix, not just the GPU's L2 cache size.
- KV-cache projection (Exp I4). $k^\star$=64 on SmolLM2-135M at 32K context (15.8× VRAM savings). Measured on EC2 L40S. The compressed cache reduces memory bandwidth pressure in addition to VRAM footprint.

## 5.2 Cross-GPU Generalization (Open)

The cross-GPU prediction table in Table 1 provides analytic predictions for 10 GPU types based on the L2 cache residency model calibrated to the RTX 4070 Laptop measurement. Only the RTX 4070 Laptop row is empirically confirmed at Llama-8B scale. The remaining 9 GPU types (RTX 4090, A100, H100, L40S, A10G, L4, RTX 3080, RTX 4080, RTX 5060) await cross-GPU validation hardware. The cross-validation use (`scripts/p3_cross_gpu.py`) auto-detects GPU type and predicts $k^\star$; running it on any of the unconfirmed GPUs would provide the definitive measurement.

## 6 Generalization Beyond ML

The super-baseline condition is not specific to machine learning. Any bandwidth-bound kernel that reads a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$ and can be restructured to read a compressed representation $\mathbf{W}_k \in \mathbb{R}^{m \times k}$ plus a basis $\mathbf{P} \in \mathbb{R}^{n \times k}$ satisfies Equation (2) when:

1. $k < \min(m, n)/2$ (meaningful dimensionality reduction),
2. $S(k) \leq 0.75\, C_{\mathrm{L2}}$ (working set fits in cache),
3. The kernel can be fused (reducing launch overhead).

Candidate non-ML workloads include:

- Database compression. Column-oriented databases store tables as matrices. Compressing infrequently-accessed columns via PCA and fusing the decompression into the query engine could accelerate full-table scans.
- Signal processing. FIR/IIR filter banks with many taps can be compressed via basis projection of the filter coefficient matrix. Fused filter-apply kernels could reduce memory traffic.
- Graphics. Texture and material parameter matrices in rendering pipelines could be compressed and decompressed on-the-fly in fused shader passes.

These are speculative extensions; the framework in § 2 provides the condition to test them.

## 7 Limitations

- Only one GPU measurement exists (RTX 4070 Laptop). All other rows in Table 1 are predictions.
- The byte-ratio formula assumes pure bandwidth-bound regime. On compute-bound GPUs (H100 at large batch), the FLOP overhead may dominate and the super-baseline may not appear.
- Kernel classes beyond attention are identified structurally but not measured. LoRA, KV-cache, and MoE routing remain predictions.
- Quantisation level (Q4_K_M) affects the byte ratio. FP16 weights would shift $k^\star$ upward (more bytes to save).

## 8 Reproduction

```python
# GPU predictions for all 10 GPU types
python scripts/super_baseline_general.py --all-gpus

# Kernel class analysis
python scripts/super_baseline_general.py --kernels

# Cross-GPU validation (on any CUDA GPU)
python scripts/p3_cross_gpu.py \
  --model models/llama3.1-8b-instruct-q4_k_m.gguf \
  --ranks 512,768,1024,1280,1536,2048
```

## References


[1] W. K. O. Stewart. "Geodesic Runtime Compression and the 106% Baseline Anomaly". In: *HyperTensor Part I, to appear* (2026).

[2] W. K. O. Stewart. "Composing Compression with Geodesic Speculative Decoding". In: *HyperTensor Part III, to appear* (2026).